

CHAPTER 7

Haloalkanes and Haloarenes

Classification of Organohalogen Compounds, Nature of C-X Bond

1. The correct sequence of bond enthalpy of 'C-X' bond is: (2021)

- $\text{CH}_3 - \text{F} > \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
- $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
- $\text{CH}_3 - \text{Cl} > \text{CH}_3 - \text{F} > \text{CH}_3 - \text{Br} > \text{CH}_3 - \text{I}$
- $\text{CH}_3 - \text{F} < \text{CH}_3 - \text{Cl} < \text{CH}_3 - \text{Br} < \text{CH}_3 - \text{I}$

Haloalkanes and Chemical Reactions of Haloalkanes

2. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on? (2021)

- Hund's rule
- Hofmann rule
- Huckel's rule
- Saytzeff's rule

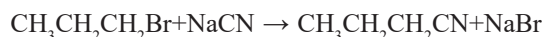
3. Elimination reaction of 2-Bromo-pentane to form pent-2-ene is (2020)

- β -Elimination reaction
 - Follows Zaitsev rule
 - Dehydrohalogenation reaction
 - Dehydration reaction
- (1), (3), (4)
 - (2), (3), (4)
 - (1), (2), (4)
 - (1), (2), (3)

4. An example of a sigma bonded organometallic compound is: (2017-Delhi)

- Cobaltocene
- Ruthenocene
- Grignard's
- Ferrocene

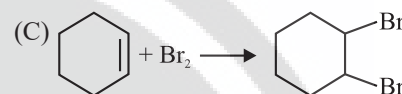
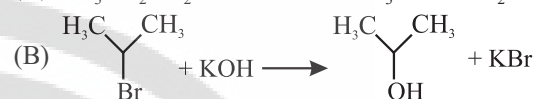
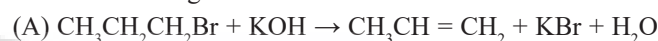
5. Consider the reaction



This reaction will be the fastest in: (2016 - II)

- N,N'-dimethylformamide (DMF)
- Water
- Ethanol
- Methanol

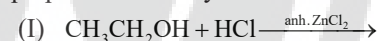
6. For the following reaction:



Which of the following statements is correct? (2016 - I)

- (A) is substitution, (B) and (C) are addition reactions.
- (A) and (B) are elimination reactions and (C) is addition reaction.
- (A) is elimination, (B) is substitution and (C) is addition reaction.
- (A) is elimination, (B) and (C) are substitution reactions.

7. Which of the following reaction(s) can be used for the preparation of alkyl halides? (2015 Re)



- (III) and (IV) only
- (I), (III) and (IV) only
- (I) and (II) only
- (IV) only

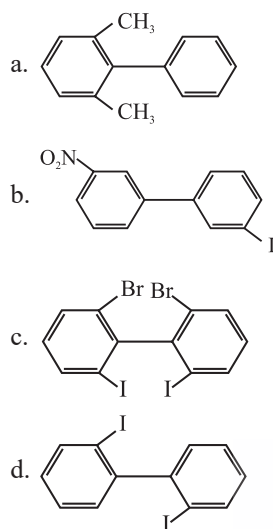
Stereochemical Aspects of Nucleophilic Substitution Reactions

8. The incorrect statement regarding chirality is: (2022)

- A racemic mixture shows zero optical rotation.
- $\text{S}_{\text{N}}1$ reaction yields 1 : 1 mixture of both enantiomers.
- The product obtained by $\text{S}_{\text{N}}2$ reaction of haloalkane having chirality at the reactive site shows inversion of configuration.
- Enantiomers are superimposable mirror images on each other.

9. Which of the following biphenyls is optically active?

(2016 - I)



10. Two possible stereo-structures of $\text{CH}_3\text{CHOHCOOH}$, which are optically active, are called:

(2015 Re)

- Mesomers
- Diastereomers
- Atropisomers
- Enantiomers

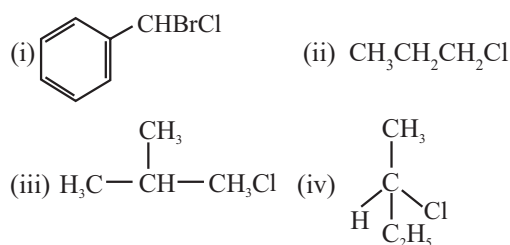
11. In a $\text{S}_\text{N}1$ reaction on chiral centres, there is:

(2015 Re)

- 100% inversion
- 100% racemisation
- Inversion more than retention leading to partial racemisation
- 100% retention

12. Which of the following compounds will undergo racemisation when solution of KOH hydrolyses?

(2014)

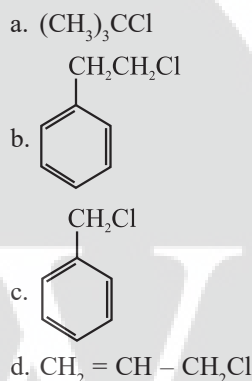


- (ii) and (iv)
- (iii) and (iv)
- (i) and (iv)
- (i) and (ii)

Haloarenes and Chemical Reactions of Haloarenes

13. Which of the following will NOT undergo $\text{S}_\text{N}1$ reaction with OH^- ?

(2020-Covid)



Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13
a	d	d	c	a	c	b	d	c	d	c	c	b